

BIWEEKLY INTERN PROGRESS REPORT

Document Intelligence & Retrieval Pipeline

Embedding model benchmarks → hybrid retrieval leaderboard → parsing re-evaluation with GLM-OCR

What this report covers

01

Week 5 Recap: Embedding Models

9 dense, sparse, multi-vector, and vision-language embedding models registered and benchmarked

02

Week 5: Hybrid Evaluation Harness

Built a 4-stage retrieval pipeline (BM25 + dense + SPLADE → RRF fusion → LLM judge) and ran it on a 54-pair QA bank

03

Week 6: Revisiting Parsing with GLM-OCR

Benchmarked a new single-model 0.9B VLM against our locked-in multi-stage parsing pipeline

04

Week 7 Plan

Formal CER/TEDS for GLM-OCR, hallucination mitigation, and the remaining embedding-model candidates

Nine models, four representation types

All backends pinned to commit SHAs in models.yaml and benchmarked on the same 54-pair ground-truth QA bank.

✓ Registry locked — 9 models spanning dense, sparse, multi-vector, and vision-language embeddings, evaluated under one harness

DENSE

qwen3-embedding-8b (4-bit)

GO-TO BACKEND

Best overall — top Spec Hit Rate (1.00) and top Faithfulness (0.454) when hybridized with BM25.

SPARSE / LEXICAL

BM25 (custom tokenizer)

GO-TO BACKEND

Custom regex tokenizer preserves full alphanumeric part numbers (e.g. 352952) — the backbone of every hybrid win.

MULTI-VECTOR

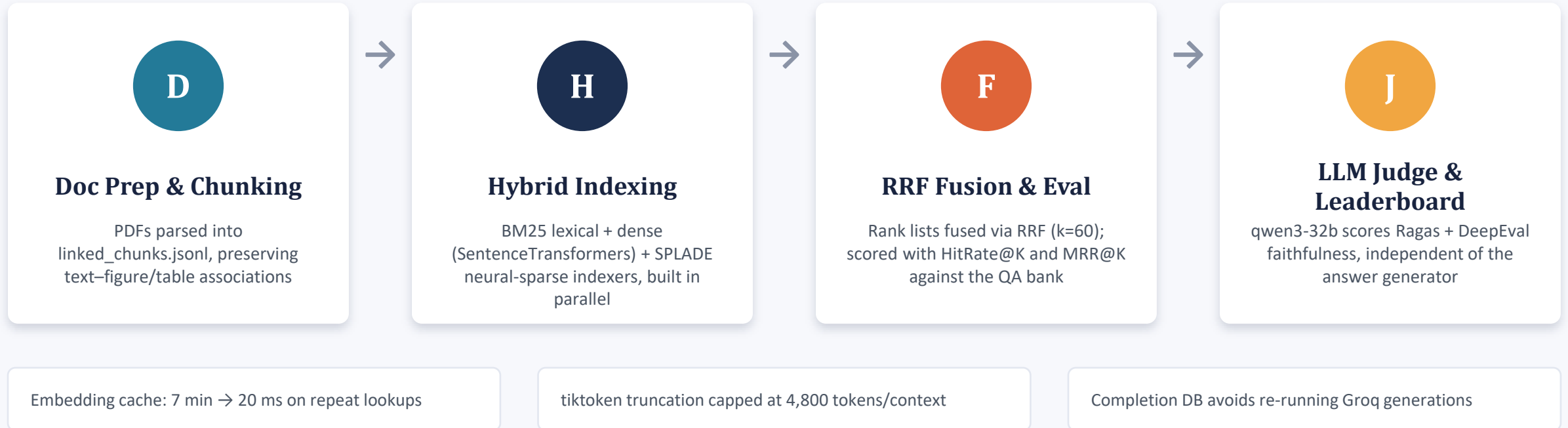
bge-m3

GO-TO BACKEND

Only model producing dense + sparse + ColBERT vectors simultaneously — broadest single-model coverage.

From documents to a leaderboard

run_leaderboard.py — four stages, fused with Reciprocal Rank Fusion (k=60)



Top model / strategy combinations

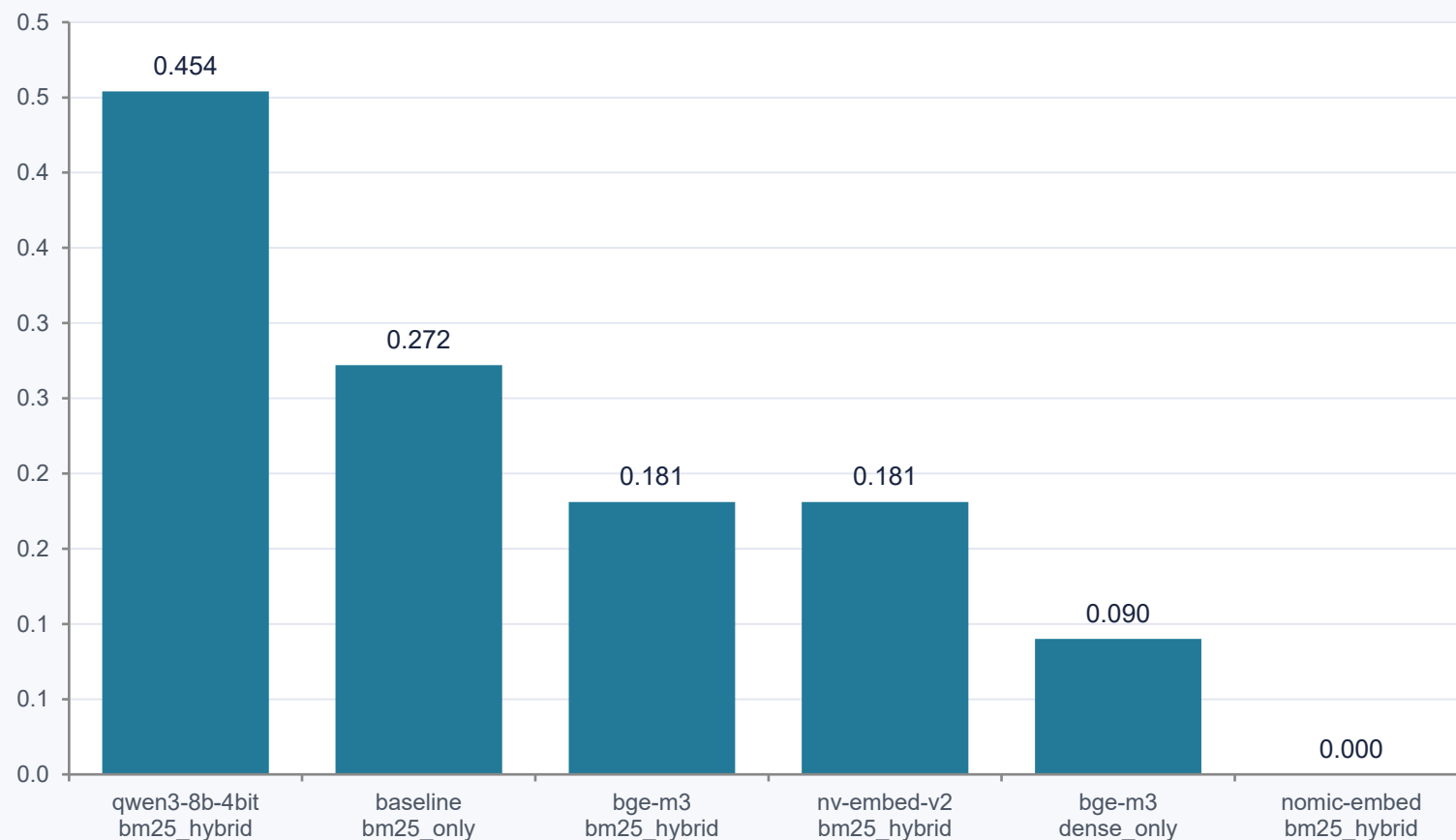
Scored across all 54 Q&A pairs — Spec Hit Rate, Overall Hit Rate, and downstream generator Faithfulness

Rank	Model	Strategy	Spec Hit@5	Overall Hit@5	Faithfulness
1	qwen3-embedding-8b-4bit	bm25_hybrid	1.00	0.518	0.454
2	baseline	bm25_only	1.00	0.722	0.272
3	bge-m3	bm25_hybrid	1.00	0.444	0.181
4	nv-embed-v2-fp16	bm25_hybrid	1.00	0.500	0.181
5	bge-m3	dense_only	1.00	0.240	0.090
6	nomic-embed-text	bm25_hybrid	1.00	0.462	0.000
7	bge-m3	dense_sparse_colbert_hybrid	0.50	0.166	0.090
8	granite-vision-embedding	text_only	0.50	0.363	0.000

qwen3-embedding-8b-4bit + bm25_hybrid ranks #1 on Faithfulness (0.454) — exact-match lexical context prevents the generator from hallucinating spec values.

Faithfulness by model / strategy

Downstream RAG-generator faithfulness (Ragas), for the six models with a perfect Spec Hit Rate@5



READING THE CHART

- All six arms hit a perfect 1.00 Spec Hit Rate@5
- Faithfulness is where they separate — hybrid BM25 arms dominate
- nomi-embed-text scores 0.000 despite perfect retrieval — a generation-side gap, not a retrieval one
- Exact-match lexical context is the strongest predictor of faithful answers

What the benchmark tells us

1.00 hit rate

Lexical BM25 wins on spec codes

Subword tokenizers fragment part/model numbers (e.g. 352952). A regex tokenizer preserving full alphanumeric strings hits 100% Hit Rate@5 on spec lookups.

RRF k=60

Hybrid fusion is essential

Neither pure dense nor pure sparse retrieval covers every query type. RRF-fused hybrid retrieval gives the best accuracy and resilience across mixed workloads.

0.454 best

Exact context → higher faithfulness

Downstream LLM faithfulness rises when exact-match lexical snippets sit in the context window, preventing the generator from hallucinating spec values.

3-way coverage

Dense, sparse & ColBERT are complementary

Dense models excel at abstract synthesis; sparse/multi-vector excel at token-level precision. bge-m3 combines all three for full coverage.

GLM-OCR: a single-model contender

zai-org/GLM-OCR (Zhipu AI) — a compact vision-language model that reads pixels and writes structured Markdown directly, no separate stages.

0.9B

PARAMETERS

vs. an 885M-param layout model alone in the old pipeline

94.62

OMNIDOCBENCH V1.5

#1 on the public leaderboard at release

1 model

REPLACES 4–5

layout + OCR + table + caption + cleanup stages, unified

INITIAL PIPELINE — 4–5 CHAINED MODELS

DocLayoutYOLO → EasyOCR / Tesseract → Docling + TableFormer/TATR → Qwen2.5VL-3B captioning → LLM cleanup pass

GLM-OCR — ONE UNIFIED VLM

Vision encoder + causal LM with Multi-Token Prediction — outputs structured Markdown (tables, headers, layout) directly from page pixels.

Text OCR: page-level results

Real GPU inference, cuda:0, GTX 1650, 4-bit quantized — scored end-to-end on full, un-cropped 300 DPI pages

Page	GT Chars	Pred Chars	CER	WER	Latency
1 (Cover)	416	478	0.8341	0.8667	105.6s
2 (Diagnostics Catalog)	1,734	1,541	0.4683	0.5882	148.9s
4 (Ordering & Contact)	1,563	1,442	0.4210	0.6468	134.3s
Filtered Average (1, 2, 4)	3,713	3,461	0.4894	0.6441	129.6s / page

Excluded: Page 3 — GLM-OCR entered a repetition loop, generating 3,831 chars of table content vs. 1,076 GT chars ($\approx 3.5\times$ repeat). See Slide 11.

Not apples-to-apples yet: baseline EasyOCR CER (0.1078) was scored on 64 pre-cropped element bounding boxes. GLM-OCR CER (0.4894) is end-to-end on full uncropped pages — it also absorbs line-ordering, header/footer, and Markdown-formatting differences the cropped baseline never sees.

Table extraction: GLM-OCR takes the lead

Evaluated with GLM-OCR's official “Table Recognition:” task prompt, which activates its structured HTML decoder head

Model	TEDS	TEDS (Structure)	Cell F1
Docling (TableFormer)	0.7295	0.7295	0.9828
TATR (Table Transformer)	0.7444	0.8684	0.4213
EasyOCR + Structure	0.9816	1.0000	0.8720
GLM-OCR (official prompt)	0.9996	1.0000	1.0000

0.9993

e0043 — Page 3

Patient Monitor Feature Comparison (14×4)

1.0000

e0074 — Page 4

ServicePlus Plans (4×3)

GLM-OCR beats every baseline table specialist — 0.9996 TEDS vs. 0.9816 (EasyOCR+Structure), 0.7444 (TATR), 0.7295 (Docling).

Findings & recommendation

What the page-level CER and prompted-TEDS results change about our read on GLM-OCR

1

Prompting Unlocks Structure

Full-page parsing with no task prompt yields plain, unformatted text. The official “Table Recognition:” prompt activates a specialized decoder head that emits clean <table> HTML with 100% cell accuracy.

2

Repetition Loop on Dense Pages

Page 3’s dense table triggered a token-repetition loop — 3× the ground-truth length. Needs standard generation stop-criteria / max-new-tokens limits before it’s production-safe.

3

Eval Methodology Still Diverges

Baseline CER is scored on 64 cropped element boxes; GLM-OCR CER is scored end-to-end on full pages. The two numbers aren’t directly comparable until we run both under one protocol.

Recommendation: GLM-OCR’s single 0.9B model already beats every specialized table extractor (0.9996 TEDS) and stays #1 on OmniDocBench (94.62). Next, hold Unlimited OCR to the same protocol before picking a production parser.

Bring Unlimited OCR into the comparison, then re-run the harness

Same protocol, one more model — then the winning parser feeds directly into the Week 5 retrieval pipeline

1

Benchmark Unlimited OCR

Run the same page-level CER/WER + official-prompt TEDS protocol on Unlimited OCR for a fair three-way comparison

2

Fix the repetition loop

Add generation stop-criteria / max-new-tokens limits to GLM-OCR before it touches dense, table-heavy pages

3

Swap the baseline in the harness

Re-run the Week 5 retrieval pipeline with GLM-OCR and Unlimited OCR replacing DocLayoutYOLO + EasyOCR + Docling

4

Compare end-to-end faithfulness

Measure whether better parsing accuracy actually lifts RAG faithfulness downstream, not just parsing-stage metrics

THANK YOU

Questions & Discussion

Next check-in: Week 7 — Unlimited OCR benchmark, and a full harness re-run with GLM-OCR + Unlimited OCR replacing the baseline parser